

Doping Stability Report -- Fe in KCoO2 at 200 C

Generated: 2026-05-30 13:12 Host: KCoO2 Dopant: Fe

Executive Summary

Fe preferentially occupies the Co layer. The formation energy difference between the two layers is +6.242 eV, strongly favouring the Co site. MD confirms this preference: on the Co site the dopant MSD plateau slope is -0.00158 Å²/ps (stable), compared to 0.08760 Å²/ps on the K site. The Co-site result is isovalent (no charge mismatch), so the formation energy is fully reliable. The K-site E_f is approximate due to a charge surplus of +2 that cannot be modelled in CHGNet.

Site	E _f (eV)	Charge	Compensation	MSD pass	Coord pass	Verdict
Co	-1.042	+0	No	PASS	PASS	STABLE
K	+5.200	+2	Yes	FAIL	PASS	MIGRATION / UNFAVORABLE

Fe at the Co layer -- STABLE

Charge situation

Fe replaces Co with no change in oxidation state (isovalent substitution, charge mismatch = 0). No charge compensation is needed. The formation energy is fully reliable.

Formation Energy (thermodynamic stability)

Formation energy E _f	-1.0422 eV	PASS	< 1.0 eV to pass
---------------------------------	------------	------	------------------

E_f = -1.042 eV. This value is strongly negative, indicating that Fe incorporation is thermodynamically very favorable at the Co site.

Molecular Dynamics Stability (200 C, 25 ps)

MSD slope (late-time)	-0.00158 Å ² /ps	PASS	< 0.005 to pass (plateau = stable)
Final MSD	0.108 Å ²	----	
Max displacement	0.789 Å	----	
Coordination (min/mean/max)	4 / 5.0 / 5	PASS	relaxed = 5, tolerance +/-1
Mean bond length (MD vs relaxed)	1.987 Å (relaxed: 1.9)	----	
Lattice volume change	+0.00 %	PASS	< +/-5% to pass

Late-time MSD slope = -0.00158 Å²/ps. The negative slope is a flat plateau -- essentially no migration detected. Final MSD = 0.108 Å²; maximum displacement from starting position = 0.79 Å.

Coordination fluctuated between 4 and 5 (mean 5.0), within the +/-1 tolerance of the relaxed value (5). This is consistent with normal thermal vibration. Mean nearest-neighbour distance during MD = 1.987 Å.

Volume change = +0.00%. The host lattice is mechanically intact.

VERDICT: STABLE

Fe is thermodynamically favorable and kinetically stable on the Co site at 200 C. All four stability criteria

Doping Stability Report -- Fe in KCoO₂ at 200 C

pass: formation energy is favorable, the dopant does not migrate during MD, its coordination environment is preserved, and the host lattice volume is unchanged.

Fe at the K layer -- MIGRATION / UNFAVORABLE

Charge situation

Fe replaces K with a charge surplus of +2. CHGNet cannot model this compensation (would require Co oxidation or electron addition). The formation energy below is approximate.

Formation Energy (thermodynamic stability)

Formation energy E_f	+5.2002 eV	FAIL	< 1.0 eV to pass
------------------------	-------------------	-------------	------------------

E_f = +5.200 eV (approximate -- charge surplus not modelled in CHGNet). This value is large and positive, indicating that Fe incorporation is thermodynamically unfavorable under equilibrium conditions at the K site.

Molecular Dynamics Stability (200 C, 25 ps)

MSD slope (late-time)	0.08760 Å²/ps	FAIL	< 0.005 to pass (plateau = stable)
Final MSD	1.185 Å²	----	
Max displacement	1.612 Å	----	
Coordination (min/mean/max)	4 / 4.7 / 6	PASS	relaxed = 5, tolerance +/-1
Mean bond length (MD vs relaxed)	1.964 Å (relaxed: 2.0)	----	
Lattice volume change	+0.00 %	PASS	< +/-5% to pass

Late-time MSD slope = 0.08760 Å²/ps. The large slope indicates significant diffusion -- the dopant is migrating. Final MSD = 1.185 Å²; maximum displacement from starting position = 1.61 Å.

Coordination fluctuated between 4 and 6 (mean 4.7), within the +/-1 tolerance of the relaxed value (5). This is consistent with normal thermal vibration. Mean nearest-neighbour distance during MD = 1.964 Å.

Volume change = +0.00%. The host lattice is mechanically intact.

VERDICT: MIGRATION / UNFAVORABLE

The formation energy is favorable but Fe migrates away from the K site during MD. The relaxed structure is not the true resting site. Inspect the MSD and trajectory to identify where the dopant ends up.

Site Preference Comparison

The Co site is preferred by 6.242 eV over the K site. Formation energy: -1.042 eV (Co) vs +5.200 eV (K).

Fe preferentially occupies the Co layer. The formation energy difference between the two layers is +6.242 eV, strongly favouring the Co site. MD confirms this preference: on the Co site the dopant MSD plateau slope is -0.00158 Å²/ps (stable), compared to 0.08760 Å²/ps on the K site. The Co-site result is isovalent (no charge mismatch), so the formation energy is fully reliable. The K-site E_f is approximate due to a charge surplus of +2 that cannot be modelled in CHGNet.

Caveats and Limitations

? CHGNet is charge-neutral.

Doping Stability Report -- Fe in KCoO₂ at 200 C

The potential does not model oxidation states, polaron formation, or electron/hole localisation. Formation energies for aliovalent dopants (charge mismatch $\neq 0$) are approximate even when structural compensation defects are included.

? No Freysoldt/Kumagai correction.

Electrostatic finite-size corrections for charged defects require the host dielectric tensor and the DFT electrostatic potential, neither of which is available from a neural-network potential. For publication-quality E_f of charged defects, follow up with DFT using doped or pymatgen-analysis-defects.

? Metal-rich reference state.

Chemical potentials are computed from elemental ground-state structures. Synthesis under oxide-rich or other conditions shifts all E_f values by additive constants; the relative ordering of sites is unaffected.

? MD timescales are short (25 ps default).

This is sufficient to assess local site stability and rapid migration, but cannot resolve slow diffusion mechanisms. A non-plateauing MSD confirms instability; a plateau does not guarantee stability on longer timescales.

? Energy rankings are more trustworthy than absolute values.

CHGNet is a universal MLIP trained on r2SCAN DFT. Formation-energy errors for transition-metal oxides are typically 50-150 meV/atom. Use relative orderings (which site is preferred?) as the primary result.